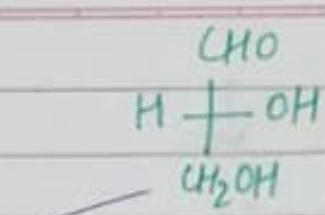
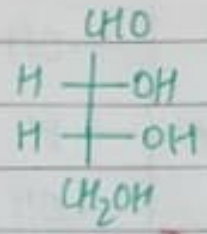
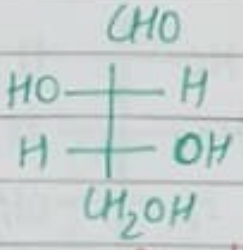
Important



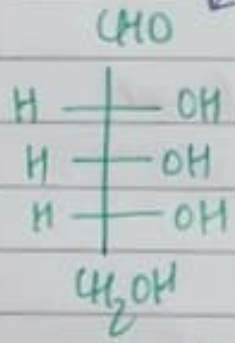
D-glyceraldehyde



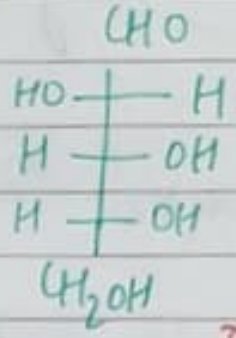
D-erythrose



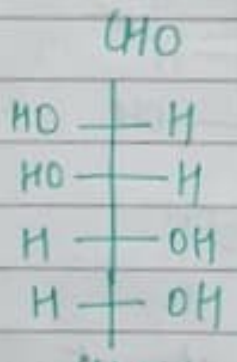
D-threose



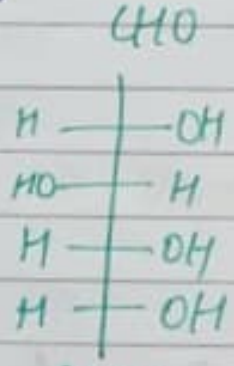
D-Ribose



D-Arabinose



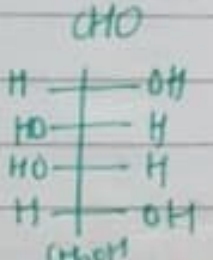
D-Mannose



D-glucose

C4 epimers

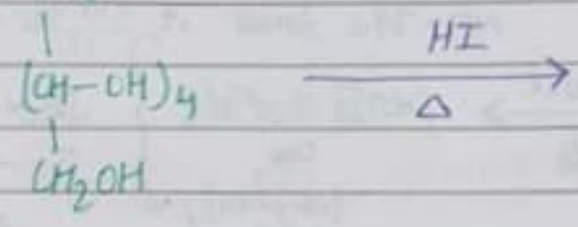
★

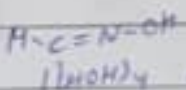


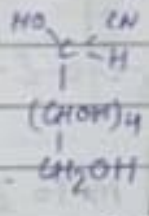
D-galactose.

Glucose

① mol. formula of glucose ⇒ $C_6H_{12}O_6$ ← from POC find empirical form. from colligative prop. ⇒ MM = 180. ⇒ mol. formula.

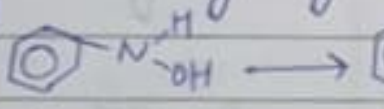
②


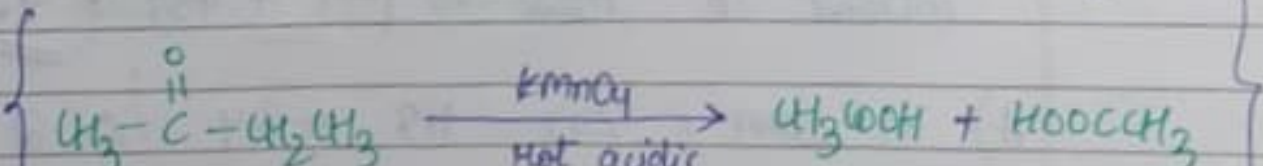
③ glucose → NH_2OH → Oxime


glucose → HCN → Cyanohydrin


} shows carbonyl compound.

④ glucose (6 'C') → Br_2, H_2O (weak O.A) → Acid (6 'C') } confirms aldehyde

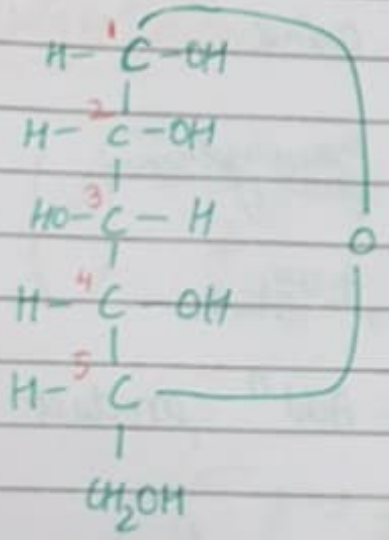
* Tollen's ⇒ ① all aldehydes ② α-hydroxy ketones ③ HCOOH ④


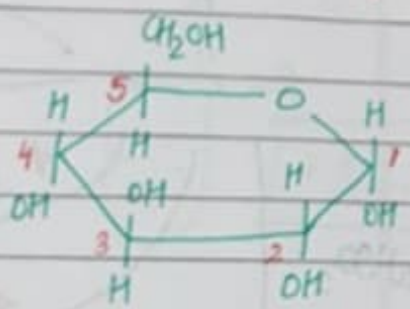
*


ketones bhi [O] hote hain lekin no. of 'C' ↓↓

(12345)
DDUD

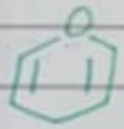
isRO pehle li
upar rakhdo



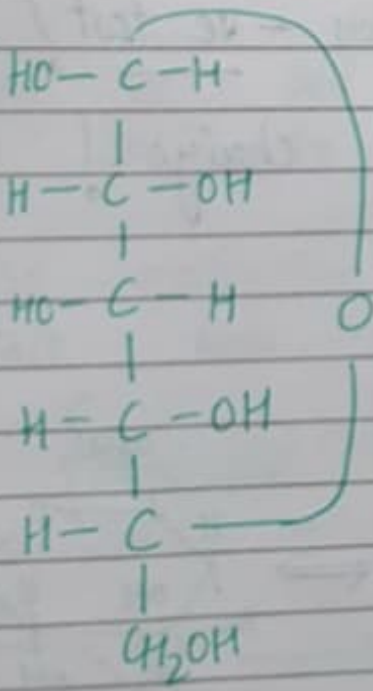


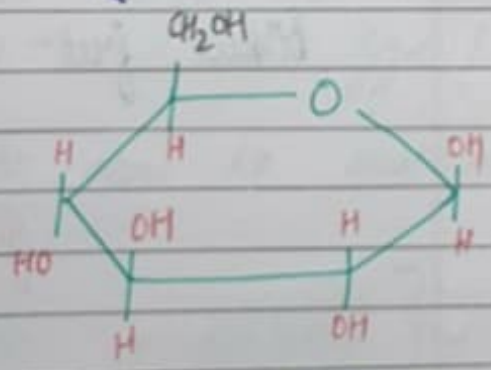
α-D-glucopyranose

Right ⇒ neeche !

(pyrane) ⇒ 

Atomers

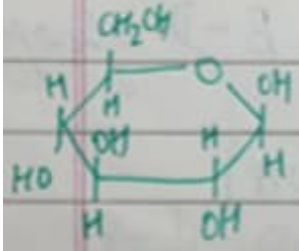


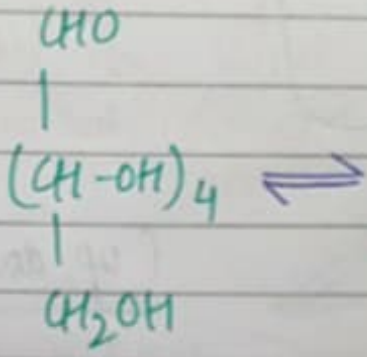


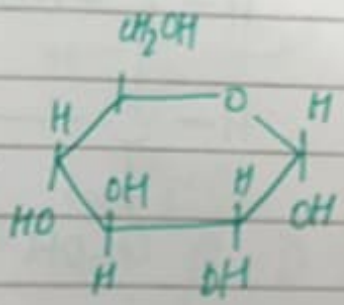
β-D-glucopyranose

(up down up down)

glucose

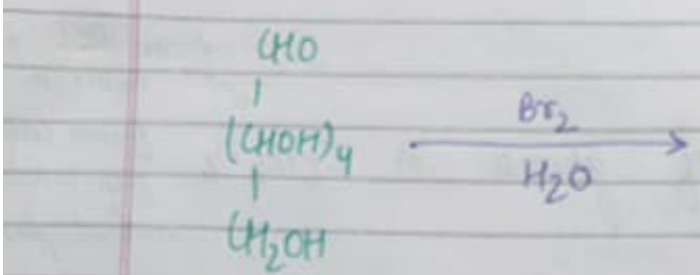






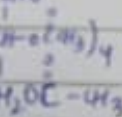
β-D | open | α-D

64% | ↓↓ very less | 36%

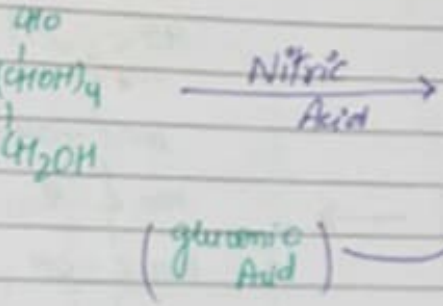


MM = 180 MM = 390 found ∴ $\frac{390-180}{42} = 5$

⑤ glucose —(acetic acid anhydride)→ penta acetate



shows 5 -OH groups on diff C atoms!

⑥ 

Glucaric acid or Saccharic acid

Reactions of Glucose : Evidence for cyclic structure

a) glucose —(Schiff's reagent)→ xxx (-ve Test!)

glucose —($NaHSO_3$)→ xxx (no add^n product)

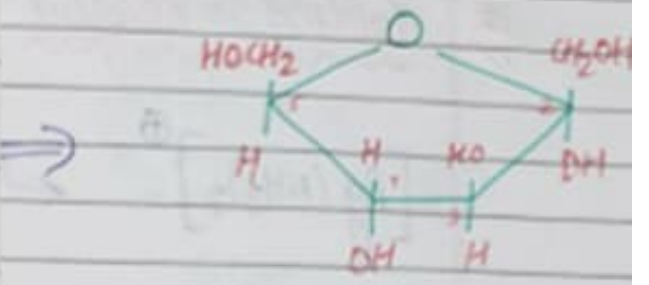
b) Pentacetate of glucose —(NH_2OH)→ xxx no rxn.

c) Glucose → when prepared in diff ways ↓ diff prop.

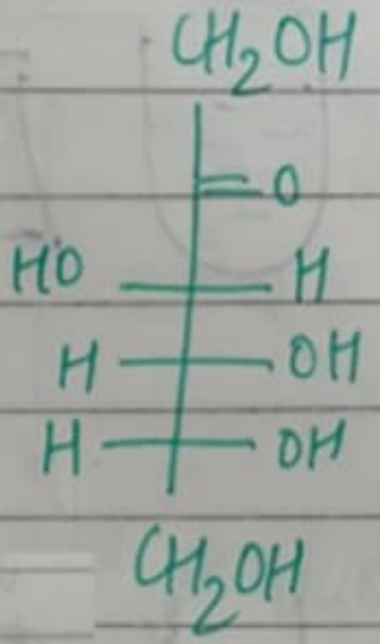
conc. sol^n of glucose —(303K, crystallize)→ glucose with T_m = 419K

glucose —(371K, cryst.)→ glucose with T_m = 423K

FRUCTOSE



α-D-fructofuranose

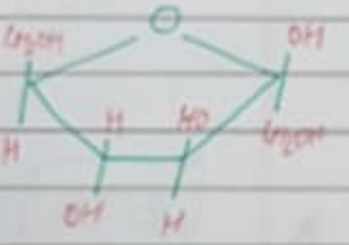


(C=O)

RRLR

D-fructose

but laevo ! (−)



β-D-fructofuranose.

D-Glucose or D-mannose or D-fructose

En-diol rearrangement $\downarrow HO^{\ominus}$

mixt of D-Glucose + D-Mannose + D-fructose

* fructose → CH_2OH | C=O | R

$\xrightarrow{HO^{\ominus}}$ CHO | $CHOH$ | R

$\xrightarrow{Tollen's}$ $[Ag(NH_3)_2]^{\oplus}$

$COO^{\ominus}$ | $CHOH$ | R

Both glucose & mannose → weak acid ✓

Less like α-Hydroxy ketones give +ve Tollen's !

Maltose

αD-gluc. αDgluc.

C_1 C_4

Reducing

(Mahendra Singh Dhoni)

Sucrose

αDgluc. βD fruc.

C_1 C_2

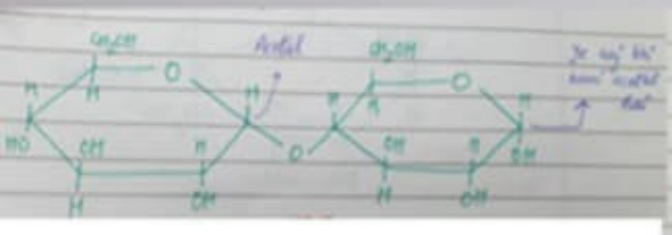
Non [R]

Lactose

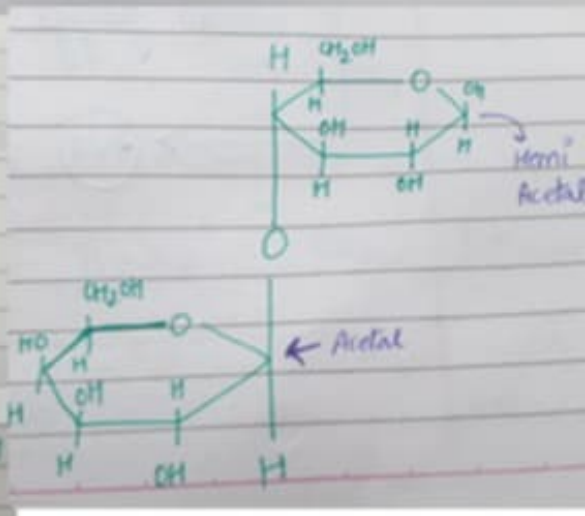
βDgalac. βDglucose.

C_1 C_4

[R]







Dissaccharide

polysaccharide

(I) Starch

Amylose 15-20%

a) water soluble
b) Unbranched chain
c) α-D Glucose
d) C_1-C_4 glyco. link

Amylopectin 80-85%

a) water insoluble
b) Branching
c) α-D Glucose
d) C_1-C_4 → chain
C_1-C_6 → branch

(II) * Glycogen → same structure like Amylopectin but more branched.

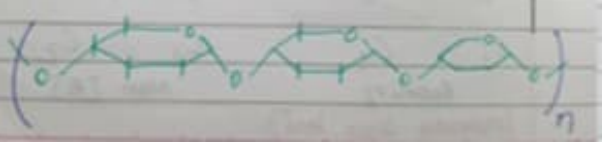
(III) Cellulose

↳ β-D Glucose!

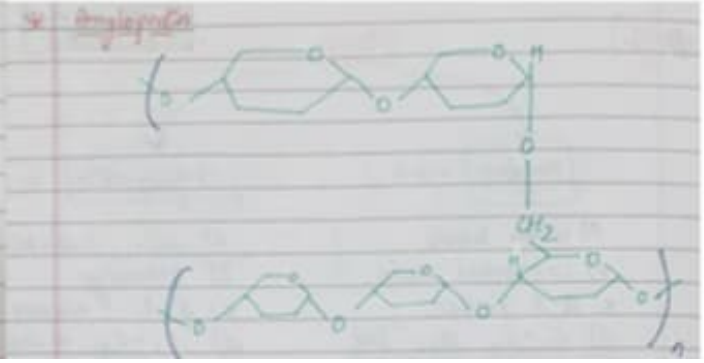
C_1-C_4 linkage!

Amylose

↳ C_1 C_4 of glucose



* Amylopectin



★★★ Imp!

TEST of CARBOHYDRATES

① Molisch Test

Molisch → Violet ✓ → Molυ

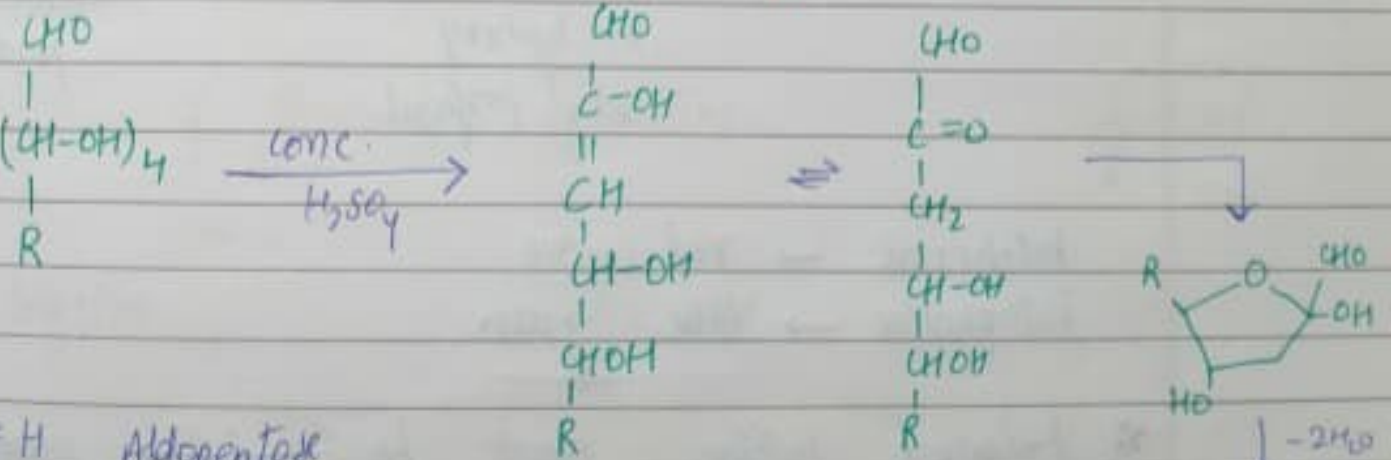
Aq. Sol^n of carb (Aldose / ketose)
+
Alcohol. Sol^n of 1-naphthol

conc. H_2SO_4 ↓

violet colour at junction

★ monosaccharides give +ve ✓

Reaction: (NOT required) ← don't learn!



R = H Aldopentose
= CH_2OH Aldohexose

R-H → furfural
R-CH_2OH → 5 hydroxy methyl furfural

1-naphthol ↓ Conc. H_2SO_4
violet

Imp:
① Furfural is involved
② Violet colour

② # Barfoed's Test:

To distinguish monosaccharides from Dissaccharides

Reagent ⇒ Cupric acetate $Cu(CH_3COO)_2$

$$RCHO + Cu^{+2} \longrightarrow RCOOH + Cu_2O$$

Monosaccharide → red

③ # Seliwanoff's Test:

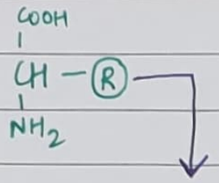
To distinguish Ketoses from Aldoses.

Ketoses $\xrightarrow{\text{Conc. } H_2SO_4}$ Furfural or 5-hydroxy methyl furfural $\xrightarrow{\text{OH, OH}}$ colored product

ketohexose → red colours
ketopentose → blue-green

★ Prolonged boiling must be ~~avoiding~~ avoided, otherwise Aldose can also give this test

AMINO ACIDS



① Glycine — H

② Alanine — CH_3

③ Valine — $-CH(CH_3)_2$

④ Leucine — $-CH_2-CH(CH_3)_2$

⑤ Isoleucine — $-CH(CH_3)-CH_2-CH_3$

⑥ Phenyl Alanine — $-CH_2-C_6H_5$

⑦ Tyrosine — $-CH_2-C_6H_4-OH$

⑧ Aspartic Acid — $-CH_2-COOH$

⑨ Glutamic Acid — $-CH_2CH_2COOH$

⑩ Asparagine — $-CH_2-CO-NH_2$

⑪ Glutamine — $-CH_2CH_2-CO-NH_2$

S A Tyrosine
Sirf aaj. Tyrosine
peelo, kal se gulabjamun
P C S G
All ⇒ non-ess.
★ Argentina zaroori hai!
Arginine ess. ✓

10 ⇒ ess.
10 ⇒ non ess.

⑫ Serine — $-CH_2-OH$

⑬ Threonine — $-CH(CH_3)-OH$

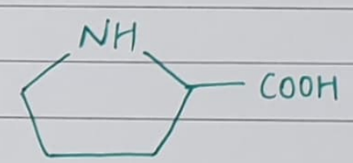
⑭ Cysteine — $-CH_2-SH$

⑮ Methionine — $-CH_2CH_2-S-CH_3$

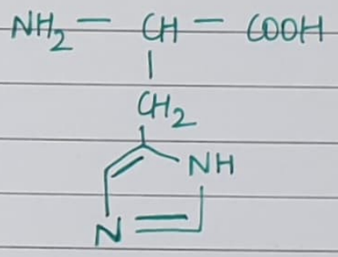
⑯ Lysine — $-CH_2CH_2CH_2CH_2NH_2$

⑰ Arginine — $-CH_2CH_2CH_2-NH-C(=NH)-NH_2$

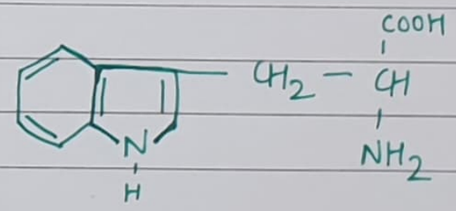
⑱ Proline

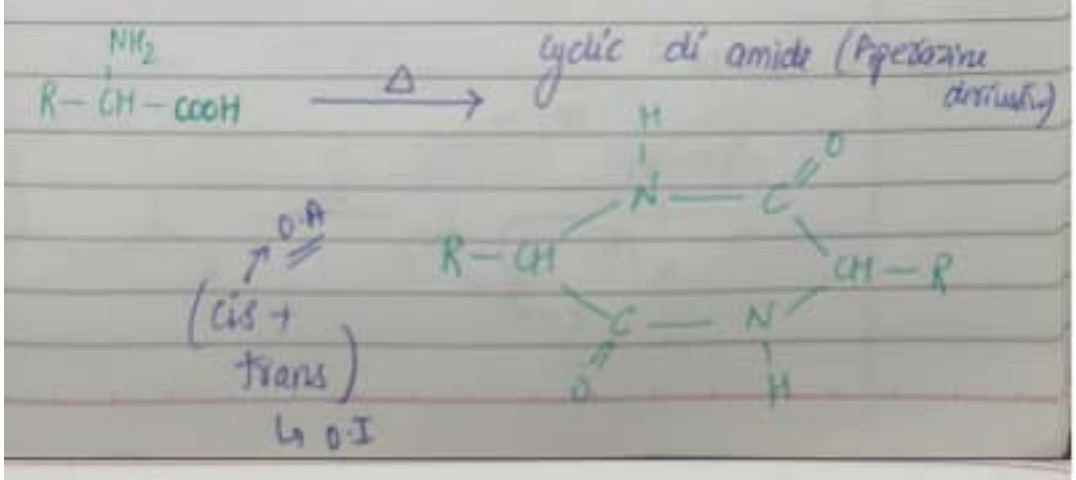


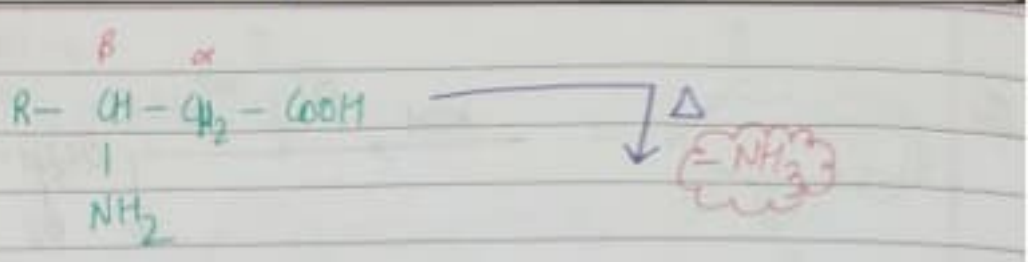
⑲ Histidine



⑳ Tryptophan







$R-CH=CH-COOH$

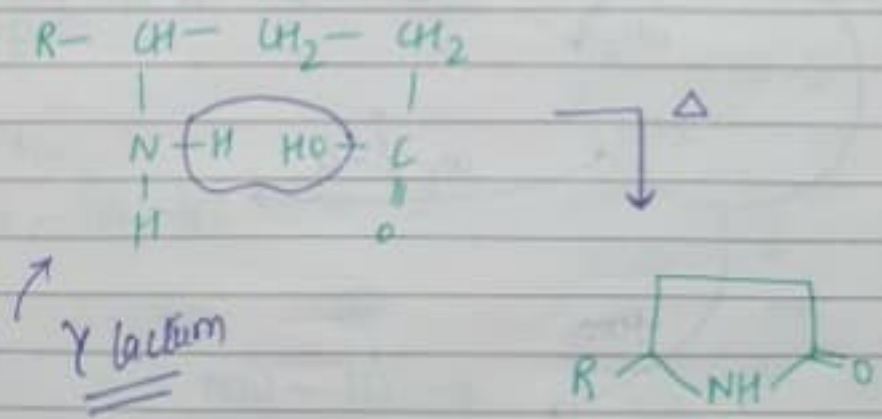
* ① $CH_2(NH_2)-CH_2COOH \xrightarrow{\Delta} CH_2=CH-COOH$

Acrylic Acid

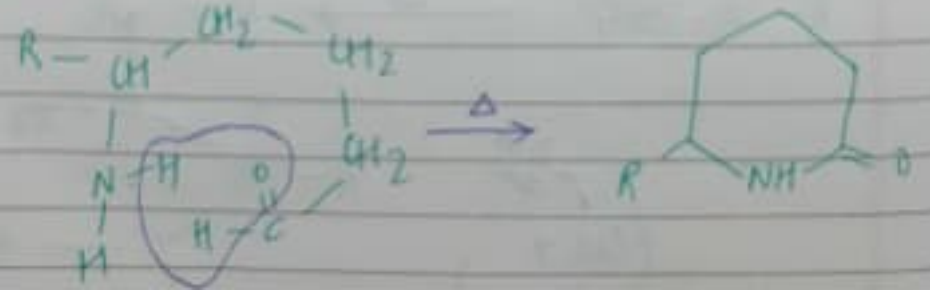
② $CH_2(NH_2)-CH-CH_2-COOH \xrightarrow{\Delta} CH_3-CH=CH-COOH$

Crotonic Acid

γ Amino Acid :



δ lactam :



pI ⇒ Isoelectric point

Neutral AAcids main pI ≈ 6

Acidic amino acids ki pI = (2.5 - 3.5)

Basic Amino acids ki pI = (9 - 10)

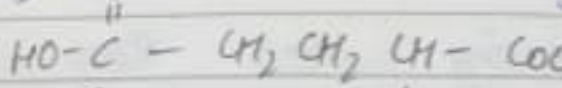


most protonated state or charge kam

max (II) ka exist karna hai
↳ matlb aisa pH chaiye jahan (II) most kam

↳ matlb agar sirf (II) hota tab jo pH uska wahi pI hai!

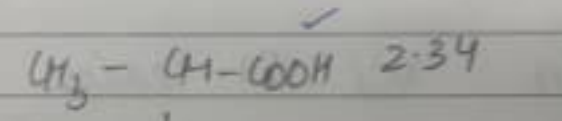
matlb ye amphiprotic kaisa hai jiska pH humein nikalna hai!

Eg.1 

2.19
4.25
9.67

Shortcut
COOH jaada hain
∴ pKa + pKa / 2
of -COOH groups

∴ 2.19 + 4.25 / 2

Eg. 

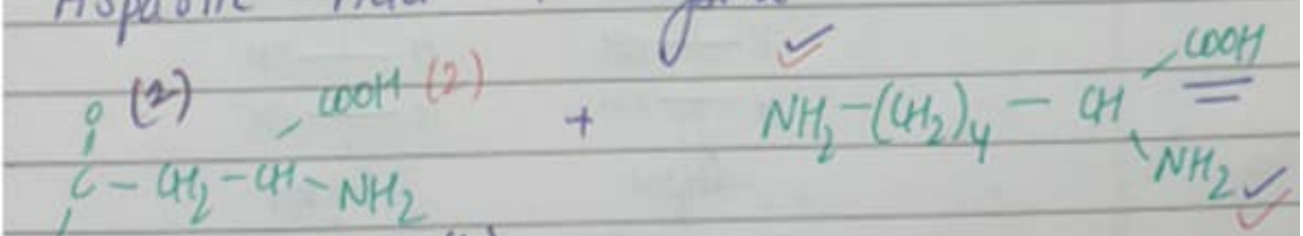
2.34
9.69

done same hain

↳ 2.34 + 9.69 / 2 = 6.02

Dipeptide

Aspartic Acid + Lysine



5 possible dipeptides.

TESTS

1. Biuret test for peptide bond

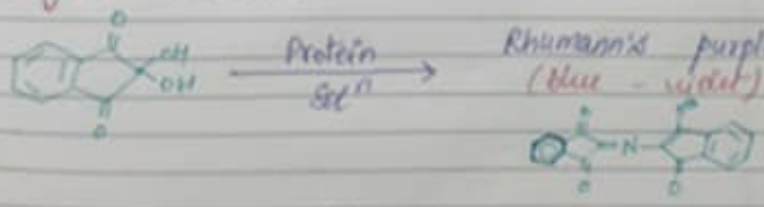
(alkaline copper sulphate + comp. containing 2 or more peptide bonds) → $[Cu(NHCONHCONH)_2]^{2-}$ violet colour

IIly Biuret → $NH_2-C(=O)-NH-C(=O)-NH_2$

also gives the same test → violet color with peptides.

In fact all comp containing → $-C(=O)-NH-C(=O)-$ also give this test.

2. Ninhydrin Reaction



Note: NH_3, RNH_2, amino acids & peptides also react with ninhydrin

3. Xanthoproteic Reaction

Aromatic ring of Amino acid or protein → (conc. HNO_3) → NaOH → orange colour

Nucleic Acids

Chemical composition :

(I) Sugar ⇒ Ribose

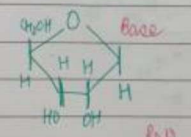
RNA ⟶ β-D-Ribose

(Hydrolysis)

DNA ⟶ β-D-2deoxy Ribose

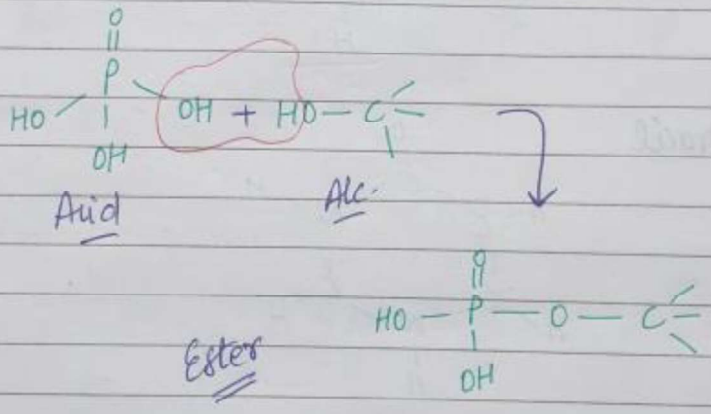
(II) # Nucleoside

pentose sugar ke (C1) pe base attach kar denge.

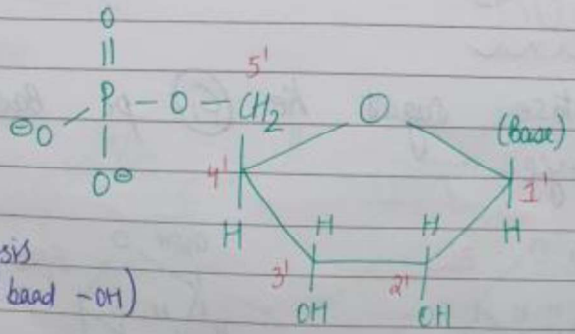


(III) Nucleotide

↳ nucleoside + phosphoric Acid ka ester C_5' of sugar.



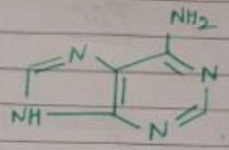
eg-



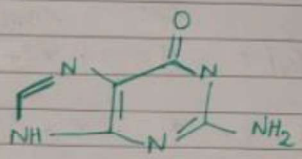
(Hydrolysis ke baad -OH)

BASES

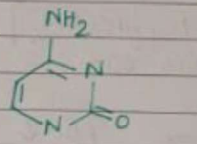
1) Adenine ⇒



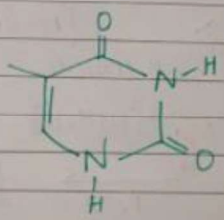
2) Guagine ⇒



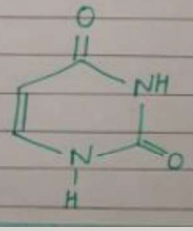
3) Cytosine ⇒

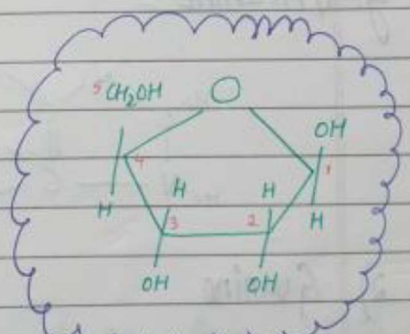


4) Thymine ⇒

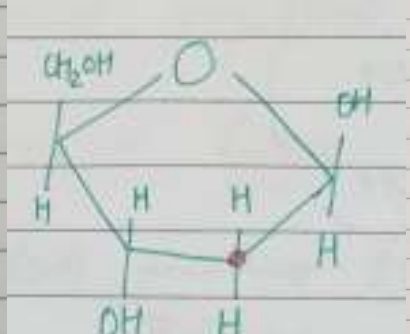


5) Uracil →

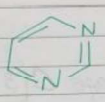




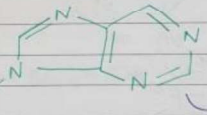
β-D-Ribose



β-D-2deoxy Ribose

*  pyrimidine

↳ {U C T} ⇒ pyrimidine derivatives.

Purine 

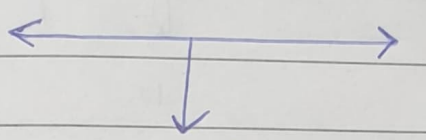
↳ {A G} ⇒ purine derivatives

Yaad karo ⇒ GAP

↳ purine

PROTEINS

① Polymer of α- amino acid

② Amino acid  Amino acid part

$-NH_2$ part

peptide linkage $\left(-NH-\overset{O}{\overset{\|}{C}}-\right)$

eg. $R-\underset{\underset{NH_2}{|}}{CH}-COOH + R'\underset{\underset{NH_2}{|}}{CH}COOH \longrightarrow R\underset{\underset{COOH}{|}}{CH}-NH-\overset{O}{\overset{\|}{C}}-\underset{\underset{NH_2}{|}}{CH}-R'$

* Dipeptide → contains 2 amino acid, 1 peptide linkage.

Tripeptide → " 3 " , 2 peptide ".

Tetrapeptide → 4 amino acid , 3 peptide "

polypeptide → ≥ 10 amino acid

Protein : polypeptide having more than 100 amino acid & molecular mass higher than 10,000.

All proteins are polypeptide but all polypeptide are not protein.

MAINS ★ Insulin is protein w/ 51 amino acids

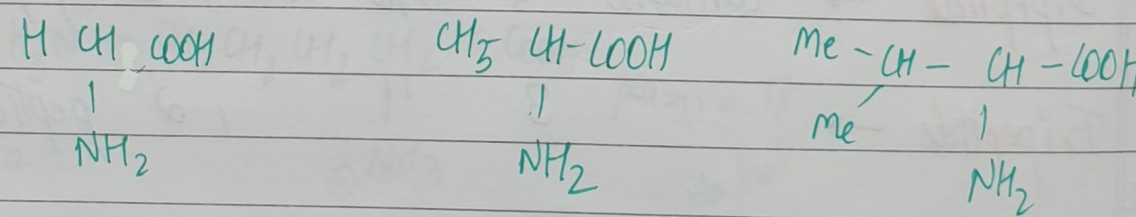
Polypeptides having less amino acids are likely to be protein if have structure like protein.

* **Structure of protein**

ADV + MAINS
(1) Primary structure of protein

MAINS only
(2) Secondary "
(3) Tertiary "
(4) Quaternary "

(I) **Primary structure** : It refers to sequence of amino acid in a protein.



Amino end → N- terminal

Carboxylic end → C- terminal

① General method of writing the sequence of amino acids in a peptide is with terminal amino group on the left.

✗ $HO_2 - A - NHCO - B - NH_2$

$H - HN - B - \overset{O}{\overset{\|}{C}} - NH - A - \overset{O}{\overset{\|}{C}} - OH$

✓ (Aise dikhate hain)

② Terminal amino group indicated by H and terminal carboxylic acid group indicated by —OH

(II) Secondary structure of Amino acids

It refers to the shape of peptide chain
a) α- helix
b) β- pleated structure.

a) α- helix

polypeptide chain forms all possible H-bonds by NH and CO group in a peptide chain and forms a spring like structure.

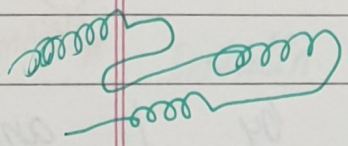
b) β- pleated structure

Peptide chain arrange side by side which are held together by H- bond. Structure similar to pleated folds of drapery (parde) (curtains)

(III) Tertiary Structure

1. It refers to 3D arrangement of polypeptide chain.

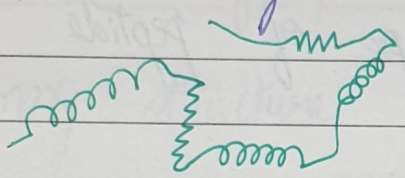
2. force - hydrogen bonds, electrostatic forces, vanderwaal forces, disulphide linkage.



(IV) Quaternary Structure

closed packed

Arrangement of the subunits of a protein that has more than one polypeptide (4, 5 tertiary structure mil gye)



MAINS only #

Denaturation of Protein

i) Protein have unique 3D structure & biological act. ⇒ native protein.

ii) Protein → (Physical / chemical change) → Loss of biological act.

Physical change ⇒ ① Temp change ② pH change. ← chemical change

known as Denaturation of protein.

* iii) During denaturation, secondary & tertiary structure gets destroyed but 1° structure remains intact.

eg. → 1) Milk $\xrightarrow[\text{juice}]{\text{lemon}}$ Cheese
2) Curdling of milk
3) Boiling of egg

Reverse process ⇩ Renaturation

Denaturation ⇒ may be REV/IRR.

MAINS only

Vitamins

Organic comp which perform specific biological fnc.

Classification:

① Fat Soluble:
1. Soluble in fat & oils but insoluble in H_2O
2. Stored in liver and adipose tissue.

eg. Vitamin: ADEK

② Water soluble:
1. Soluble in water
2. Can't be stored in our body.
3. eg. Vitamin - C, B group (B_1, B_2, B_6, B_{12})

★ (B_{12}) → get stored in body.

★ → Vitamin H: neither H_2O nor fat soluble (Biotin)

Deficiency & Diseases ⇒ Refer NCERT.

B_1: Thiamine
B_2: Riboflavin
B_6: Pyridoxine
C: Ascorbic Acid